T97 fixed-input replay: P0m hand layer + crease cut + protrusion re-attach + hand freeze 80 mm vs P0m hand layer vs no feature

Hand-layer rule	Remaining target voxels (band-excl.) request median	Median of 9 run medians	Per-request diff vs no feature: median (lower/equal/higher)	Remaining: visible / hidden (sum, 78 req.)	Finger-contact region remaining (median frac.)	Freed vs no feature (all requests)	Freed by GT class: apple / fruit / table / crate / robot / none	Fruit+table+crate share: all requests (window only)
Original closed-loop record (replay noise floor)	0.1406	0.1431	-0.0001 (39/12/27)	39 / 3824	0.0000	471081	2149 / 205 / 6762 / 10947 / 72 / 113125	3.8 % (5.2 %)
No feature (surface extraction only)	0.1325	0.1371	reference	49 / 3796	0.0500	reference	-	-
P0m hand layer (hull fill + contact allowance 10 mm)	0.1153	0.1151	-0.0084 (51/27/0)	5 / 3165	0.0000	9037	6354 / 2015 / 0 / 0 / 204 / 464	22.3 % (35.2 %)
P0m hand layer + crease cut + protrusion re-attach + hand freeze 80 mm	0.1346	0.1343	+0.0084 (26/5/47)	1 / 3716	0.0436	7701	6988 / 67 / 0 / 0 / 205 / 441	0.9 % (1.4 %)

Input: T87 fixed-input replay (B0 closed-loop k0 requests), 9 runs (hurdle 211 excluded), 435 requests, 78 grasp-window requests with a hand layer in every arm.

Definitions unchanged from outputs/2026-11-03_T87_.../재생_고정입력/README_definitions.md: grasp window = before GT grasp and palm-apple <= 150 mm; band = GT apple bottom + 6 mm excluded.

Visible = voxel holds a GT apple sample visible in the grasping-hand wrist camera at that request. Freed = no-feature <= 0 and arm > 0, MuJoCo GT class (nearest surface <= 8 mm).

No-feature and P0m hand-layer rows are the T96 replay outputs (same chunks, symlinked). Criteria for T97 are in t97-replay-criteria.